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EVEN END SEMESTER EXAMINATION June 2025

NAME OF THE COURSE : B.Tech (Civil Engg.)
NAME OF THE PAPER : Reinforced Cement Concrete - II
TIME: 3 Hours

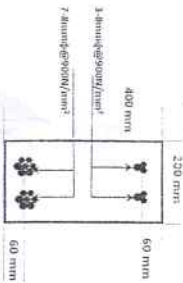
SEMESTER: VI
PAPER CODE: TCE 602
MAX MARKS: 100

Note:-

- All questions are compulsory.
- Answer any two sub questions among a, b & c in each main question, each carrying 10 marks.
- Use of IS-456:2000 supplied by the department is allowed for use in the examination.

Q1

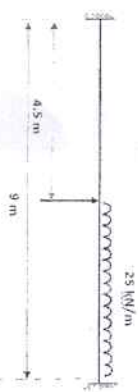
- Illustrate the various methods/approaches of the PSC beam section analysis? (2X10=20 marks)
- Determine the eccentricity of the cable force at transfer state for the given PSC flexural member section? (10)



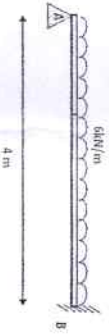
- Analyze the given prestressed fixed for the maximum critical B.M section? The cross section of the beam is rectangular with 230mm width and 350mm depth, and with a uniform cable force of 950KN provided at 75mm from the soffit of the beam. (10)



- Present the significance of shift of point of contraflexure, when limit analysis is applied to develop the bending moment envelope for a statically indeterminate beam? (2X10=20 marks)
- Develop the BM Envelope for the given condition of the continuous Beam with a 20% moment redistribution applied to the beam? (10)



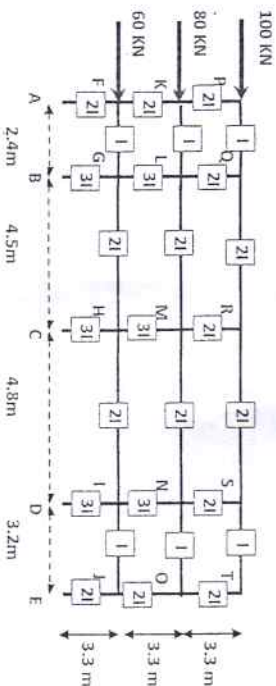
- Develop the BM Envelope and determine the shift of contraflexure at the fixed support with 30% moment redistribution? (10)



Q3

- List and explain the guidelines that should be verified while selecting a substitute frame for a given building frame. Provide reasoning for each guideline to justify its importance in the design process? (2X10=20 marks)

- Analyze the Given 2-D Building Frame and determine the approximate frame centroid position from Node A? (10)



- "Portal and Cantilever methods will give the same results while analyzing a building frame". Is the statement True or False, support with answer illustrating with a simple appropriate example & reasonings? (10)

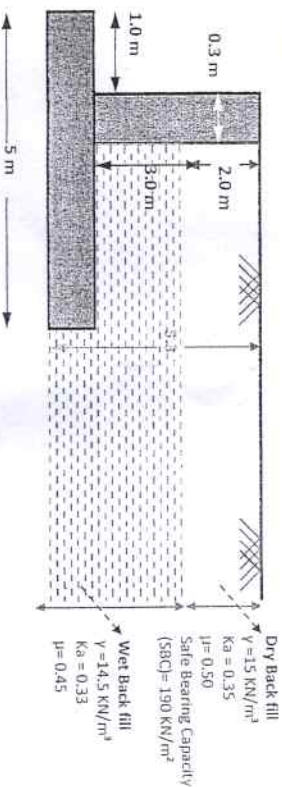
Q4

- Derive rough equation for determination of the base width of a typical cantilever retaining wall from its stability aspects? (2X10=20 marks)

- Design the Stem Wall section for the given data, and give the reinforcement detailing for the same? (10)



- Determine the magnitude and the position of the active earth pressure acting on the stem for the given retaining wall section. (10)



Q5

(2X10=20 marks)

- (a) Write short notes on the various components and their functionalities and advantages of a typical Inze Tank with the help of neat diagrams? (10+5)
- (b) Illustrate the typical analysis approach of a rectangular water tank wall panel, for On Ground and Under-Ground storage conditions? (10+5)
- (c) Propose a design for an underground rectangular water sump with a breadth restriction of 1.2m, to store water required for 4 days or above for a family of 8? (10+5)

Table: For general LSM Design Use

Stress-Strain Values for $F_y=500$				Stress-Strain Values for $F_y=415$			
K	Strain	Ultimate Stress	Idealized Curve	Corresponding Strain	K	Strain	Ultimate Stress
0.0000	0.0000	0.0000	0.00	0.000000	0.0000	0.0000	0.00
0.8000	0.0001	400.0000	347.83	0.001839	0.8000	0.0001	332.0000
0.8500	0.0003	425.0000	369.57	0.002148	0.8500	0.0003	352.7500
0.9000	0.0007	450.0000	391.30	0.002657	0.9000	0.0007	373.5000
0.9500	0.0010	475.0000	413.04	0.003065	0.9500	0.0010	394.2500
0.9750	0.0020	487.5000	423.91	0.004120	0.9750	0.0020	404.6250
1.0000	0.0035	500.0000	434.78	0.005674	1.0000	0.0035	415.0000
							360.87
							0.005304